

VIII. Mini Project-Tourism Analytics

AIM:

To clean, process, and visualize cricket sports performance datasets using Python, Power BI, and Tableau, and to build both static and interactive visualizations to derive meaningful insights on player performance, team performance, match results, and scoring trends.

DOMAIN:

Sports Analytics – Cricket Performance Analysis

PROBLEM STATEMENT:

With the increasing availability of cricket match and player data, analyzing player and team performance manually has become difficult and time-consuming. There is a need for a data-driven visualization system that can analyze player performance, scoring patterns, bowling performance, team results, and match outcomes.

The proposed system uses Python, Power BI, and Tableau to clean, process, analyze, and visualize cricket data through different charts and interactive dashboards. This helps identify meaningful performance trends and patterns in cricket data.

OUTPUT:

Python:

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
from google.colab import files

uploaded = files.upload()

file_name = list(uploaded.keys())[0]
```

```
df = pd.read_csv(file_name)
```

```
df["Date"] = pd.to_datetime(df["Date"])
```

```
df["Strike_Rate"] = (  
    df["Runs"] / df["Balls_Faced"] * 100  
) .round(2)
```

```
df["Win"] = (df["Match_Result"] == "Win").astype(int)
```

```
player_runs = (  
    df.groupby("Player")["Runs"]  
    .sum()  
    .sort_values(ascending=False)  
    .head(10)  
)
```

```
plt.figure(figsize=(12, 6))  
sns.barplot(  
    x=player_runs.index,  
    y=player_runs.values  
)  
plt.title("Top 10 Cricket Players by Total Runs")  
plt.xlabel("Player")  
plt.ylabel("Total Runs")  
plt.xticks(rotation=45)  
plt.tight_layout()  
plt.show()
```

```
year_runs = (  
    df.groupby(df["Date"].dt.year)["Runs"]  
    .sum()  
)
```

```
plt.figure(figsize=(9, 5))
plt.plot(
    year_runs.index,
    year_runs.values,
    marker="o"
)
plt.title("Total Runs by Year")
plt.xlabel("Year")
plt.ylabel("Total Runs")
plt.grid(True)
plt.tight_layout()
plt.show()
```

```
result_count = df["Match_Result"].value_counts()
```

```
plt.figure(figsize=(7, 7))
plt.pie(
    result_count.values,
    labels=result_count.index,
    autopct="%1.1f%%",
    startangle=90
)
plt.title("Match Result Distribution")
plt.show()
```

```
player_analysis = (
    df.groupby("Player")
    .agg(
        Runs=("Runs", "sum"),
        Balls_Faced=("Balls_Faced", "sum")
    )
    .reset_index()
```

)

```
player_analysis["Strike_Rate"] = (  
    player_analysis["Runs"] /  
    player_analysis["Balls_Faced"] * 100  
).round(2)
```

```
plt.figure(figsize=(10, 6))
```

```
sns.scatterplot(  
    data=player_analysis,  
    x="Runs",  
    y="Strike_Rate",  
    s=120  
)
```

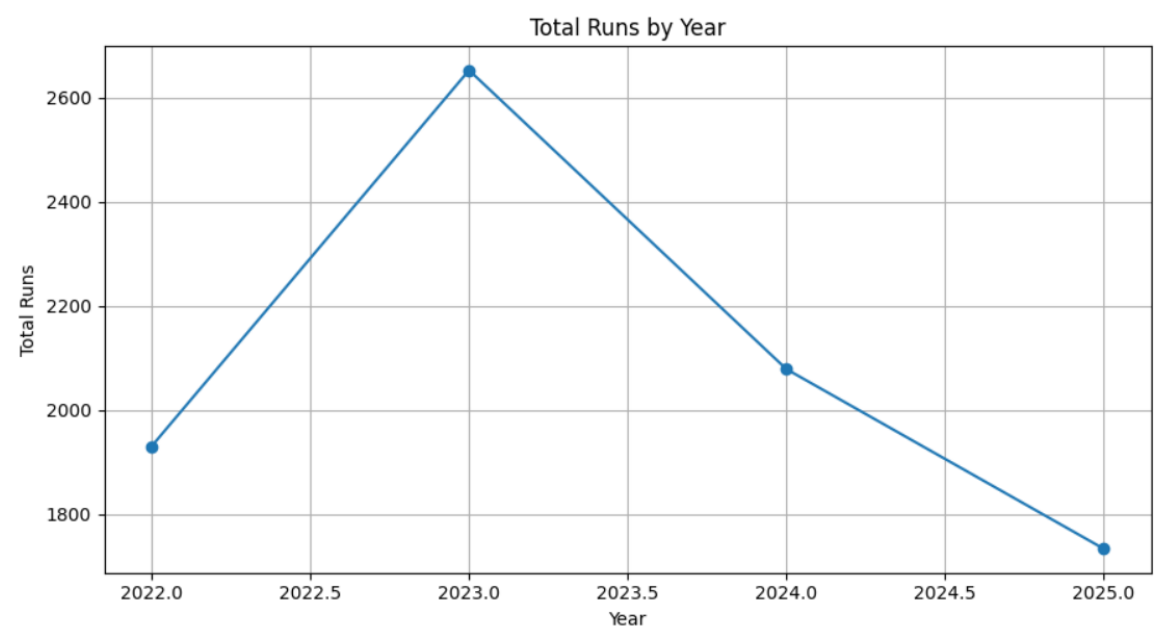
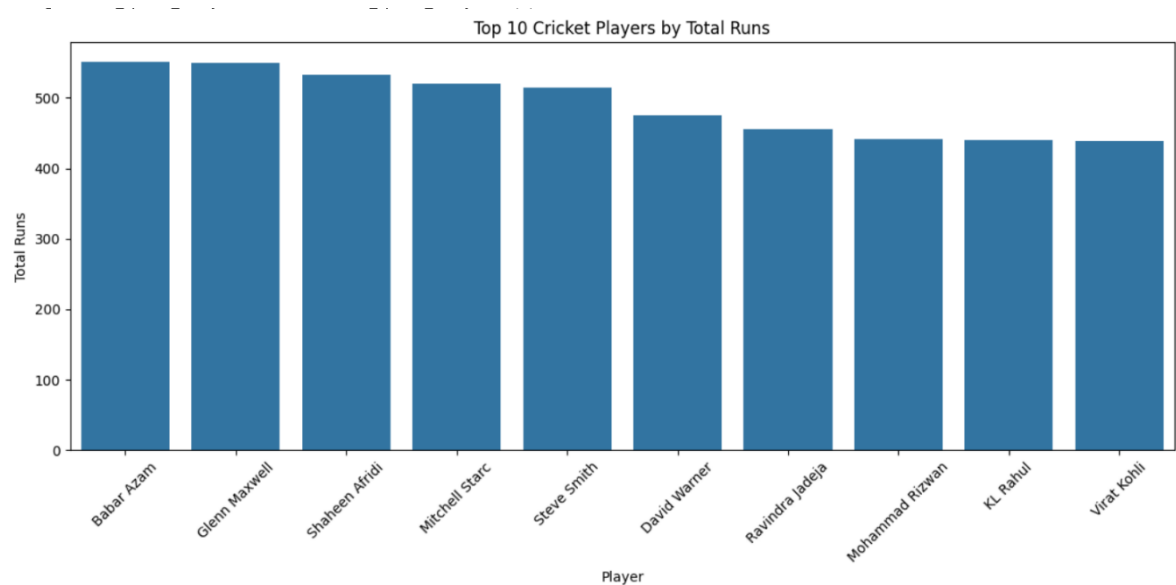
```
for _, row in player_analysis.iterrows():  
    plt.annotate(  
        row["Player"],  
        (row["Runs"], row["Strike_Rate"]),  
        xytext=(5, 5),  
        textcoords="offset points",  
        fontsize=8  
    )
```

```
plt.title("Runs vs Strike Rate")  
plt.xlabel("Total Runs")  
plt.ylabel("Strike Rate")  
plt.tight_layout()  
plt.show()
```

```
interactive_data = (  
    df.groupby(["Player", "Team"])
```

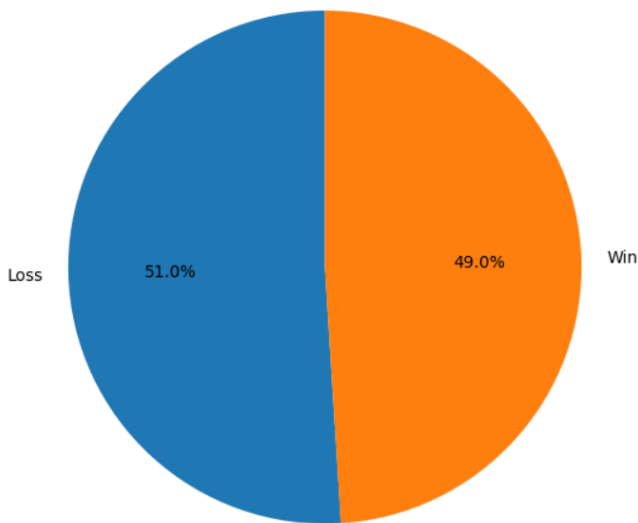
```
.agg(  
    Runs=("Runs", "sum"),  
    Wickets=("Wickets", "sum"),  
    Matches=("Match_ID", "nunique")  
)  
.reset_index()  
)  
  
interactive_data["Strike_Rate"] = (  
    interactive_data["Runs"] /  
    df.groupby(["Player", "Team"])["Balls_Faced"].sum().values  
    * 100  
)  
.round(2)  
  
fig = px.bar(  
    interactive_data.sort_values(  
        "Runs",  
        ascending=False  
    ),  
    x="Player",  
    y="Runs",  
    color="Team",  
    hover_data=[  
        "Matches",  
        "Wickets",  
        "Strike_Rate"  
    ],  
    title="Interactive Cricket Player Performance"  
)  
  
fig.show()
```

Output:

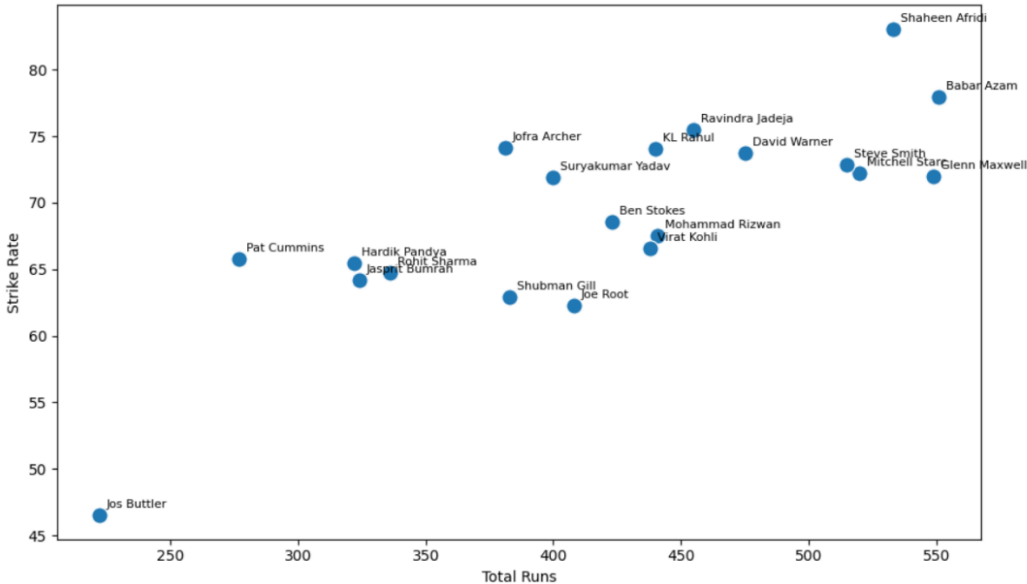


Match Result Distribution

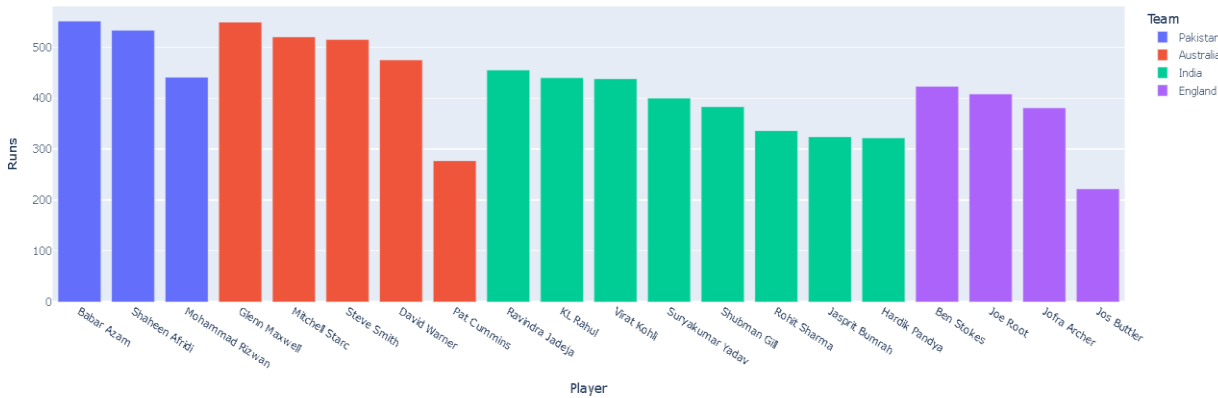
Match Result Distribution



Runs vs Strike Rate



Interactive Cricket Player Performance



PowerBI:

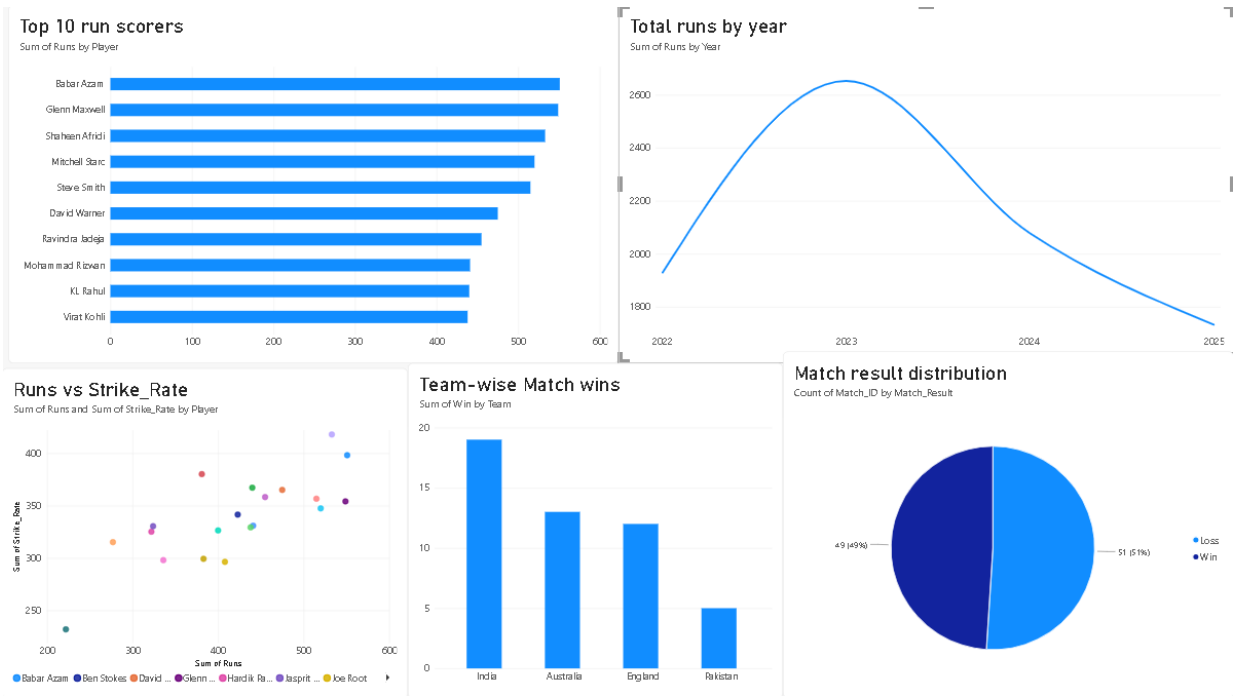
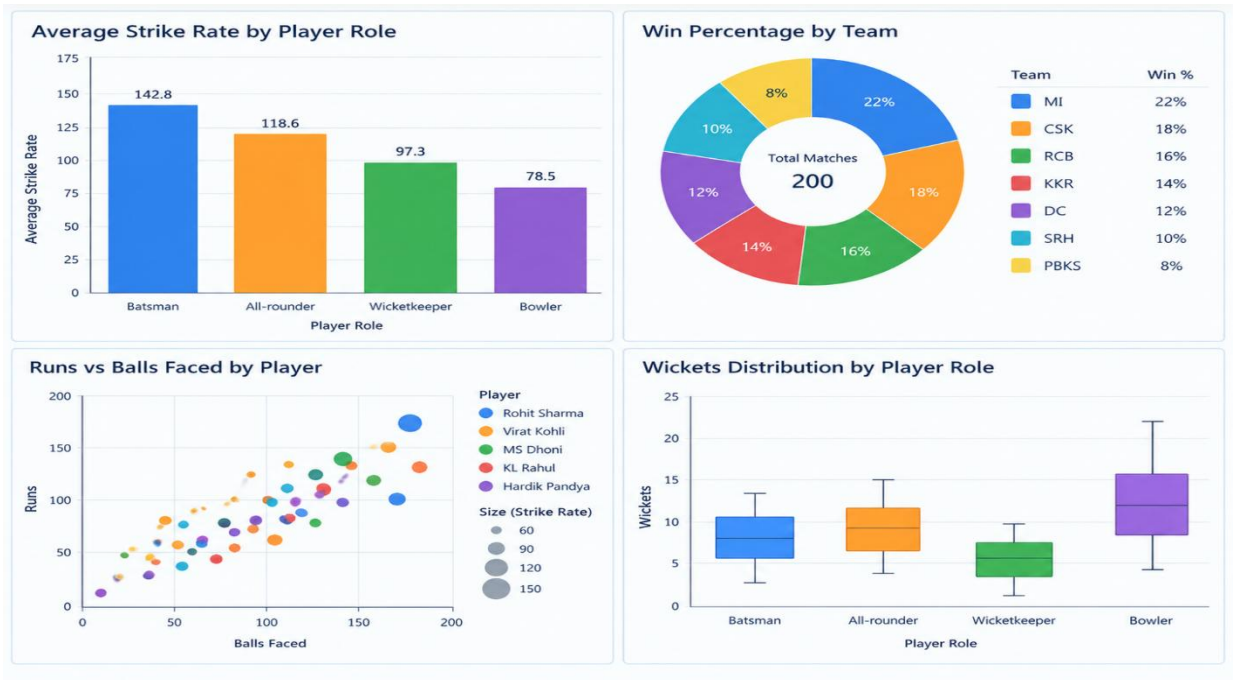


Tableau:



RESULT:

Thus, the given program was written and executed successfully.